

Question Paper

Exam Date & Time: 20-Nov-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
SEVENTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

Analog IC design [ECE_4408]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Explain with a neat diagram the drain-induced barrier lowering (DIBL) effect in MOSFET and discuss its impact on device performance. (5)

A)

Course Outcome: "CO1" / Cognitive Level: "Understand"

- B) Explain with a neat diagram the velocity saturation effect in MOSFET (5)

Course Outcome: "CO1" / Cognitive Level: "Understand"

- 2) Design the common-gate stage for the Figure 2A with following parameters: $v_{out}/v_{in} = 5$, $R_S = 0$, $R_3 = 500 \text{ ohm}$, $1/g_m = 50$, power budget = 2 mW, $V_{DD} = 1.8 \text{ V}$. Assume $\mu_n C_{ox} = 100 \mu\text{A/V}^2$, $V_{TH} = 0.5 \text{ V}$, and $\lambda = 0$. (5)

A)

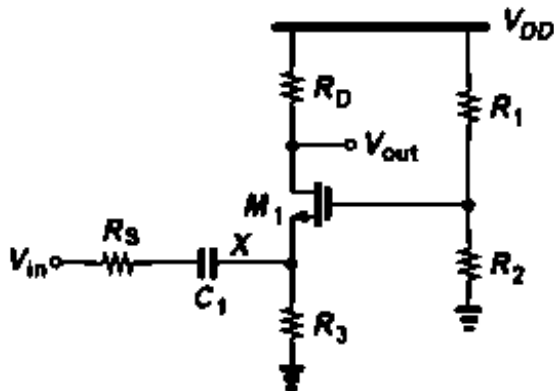


Fig 2A

Course Outcome: "CO2" / Cognitive Level: "Analysis"

- B) (5)

An integrated circuit employs the source follower and the common-source stage shown in Figure below. Design a current mirror that produces I_1 and I_2 from a 0.3-mA reference.

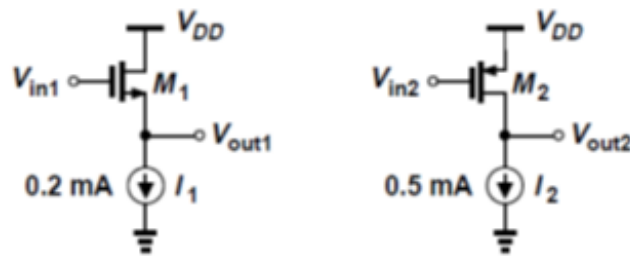


Fig 2B

Course Outcome: "CO2" / Cognitive Level: "Analysis"

- 3) Analyze the small-signal behavior of a MOS differential pair. Explain the concept of virtual ground and derive the expression for the differential voltage gain. (5)

A)

Course Outcome: "CO3" / Cognitive Level: "Analysis"

B)

- Find the range of the common mode voltage for the circuit given in the fig 3B (5)

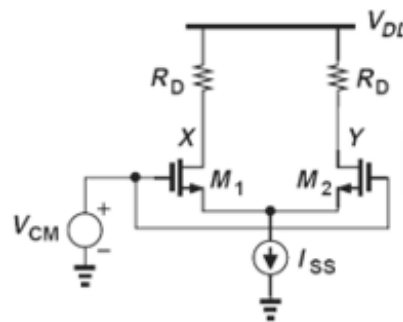


Fig 3B

Course Outcome: "CO3" / Cognitive Level: "Application"

- 4) In the amplifier of Fig.4A $R_D = 10 \text{ k}\Omega$ and $C_L = 1 \text{ pF}$. (5)

A)

Neglecting channel-length modulation and other capacitances, determine the frequency at which the gain falls by 15%? ($\sim 1.5 \text{ dB}$)

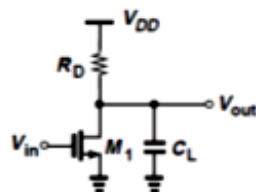


Fig 4A

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- B) An amplifier with a forward gain of A_0 and two poles at 10 MHz and 500 MHz is placed in a unity-gain feedback loop. Calculate A_0 for a phase margin of 60° (5)

- 5) Consider transistors M_1 and M_2 operating at the same drain current but with different source areas, such that the area ratio is $N = \frac{A_2}{A_1} = 10$. At a temperature of $T = 300$ K, with thermal voltage $V_T = \frac{kT}{q}$ (where $k/q = 8.617 \times 10^{-5}$ V/K), a resistor $R = 15$ k Ω is connected to produce a proportional-to-absolute-temperature (PTAT) current, given by

$$I_{PTAT} = \frac{\Delta V_{BE}}{R}.$$

Determine the values of ΔV_{BE} and I_{PTAT} at 300 K.

Course Outcome: "CO5" / Cognitive Level: "Analysis"

- B) Describe all the performance Parameters of Linear Voltage Regulators (5)

Course Outcome: "CO5" / Cognitive Level: "Application"

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Event Name	Subject
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058-END SEMSTER-ODD-2025 ECE_4408:Analog IC design

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	10
4	4	CO 4	10
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-END SEMSTER-ODD-2025 ECE_4408:Analog IC design

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	2	10(20%)
3	Application	3	15(30%)
4	Analysis	5	25(50%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		10	50 (100%)